

OSTROV KOTEL'NYJ, Russian Federation

WMO: 214320

Lat: 76.000N

Lon: 137.867E

Elev: 8

StdP: 101.23

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-38.4	-37.1	-40.5	0.1	-38.2	-39.3	0.1	-37.0	16.7	-27.2	15.0	-24.3	3.0	140	1.173

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.9	11.8	9.3	9.7	7.7	7.8	6.3	9.6	11.6	7.8	9.5	6.3	7.7	7.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.2	6.7	10.7	6.4	6.0	8.8	5.1	5.4	7.2	28.5	11.6	24.5	9.5	21.4	7.8	17.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.1	13.3	12.0	-40.0	15.3	1.3	3.6	-41.0	17.8	-41.8	19.9	-42.5	21.9	-43.5	24.6		
(5)			-40.1	11.7	1.3	3.6	-41.1	14.3	-41.9	16.4	-42.6	18.4	-43.6	21.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-13.0	-28.3	-28.7	-26.6	-18.4	-8.1	0.3	3.3	3.2	-0.1	-8.1	-19.5	-25.8		
	DBStd	13.12	4.97	5.14	5.53	5.50	4.44	3.08	2.87	3.02	2.90	5.05	5.49	5.11		
	HDD10.0	8390	1188	1083	1135	851	562	292	210	211	304	561	883	1111		
	HDD18.3	11428	1446	1316	1393	1101	820	541	467	468	554	819	1133	1369		
	CDD10.0	3	0	0	0	0	0	1	1	1	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	5.5	5.5	5.4	5.1	5.1	5.6	5.7	5.7	5.7	5.7	5.9	5.2	5.3		
(15) Precipitation	PrecAvg	137	5	5	5	5	8	17	29	24	20	13	6	4		
	PrecMax	226	21	16	20	13	24	43	66	78	61	45	17	12		
	PrecMin	72	0	0	0	0	1	3	4	3	3	1	0	0		
	PrecStd	34	4	4	4	4	6	11	15	15	11	9	4	3		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-13.3	-12.9	-7.2	-2.9	2.2	15.0	15.2	15.2	7.2	1.1	-3.6	-9.4		
		MCWB	-13.5	-13.2	-7.9	-3.3	1.0	11.7	12.6	12.1	5.8	0.5	-4.1	-10.0		
	2%	DB	-16.1	-16.4	-12.0	-6.3	0.7	8.9	11.8	11.7	5.2	0.1	-6.8	-14.1		
		MCWB	-16.4	-16.7	-12.2	-6.8	-0.1	6.8	9.2	9.3	4.4	-0.5	-7.2	-14.5		
	5%	DB	-18.4	-18.8	-16.4	-8.2	-0.6	6.0	9.7	9.7	4.2	-0.9	-9.4	-16.8		
		MCWB	-18.7	-19.1	-16.7	-8.7	-1.3	4.6	7.6	7.9	3.5	-1.5	-9.9	-17.1		
	10%	DB	-20.5	-20.8	-18.8	-10.3	-2.0	3.8	7.8	7.9	3.2	-1.8	-11.7	-18.7		
		MCWB	-20.8	-21.0	-19.0	-10.7	-2.7	2.8	6.2	6.7	2.6	-2.5	-12.1	-19.0		
	0.4%	WB	-13.6	-13.2	-7.9	-3.3	1.1	12.0	12.7	12.3	6.3	0.7	-3.9	-9.9		
		MCDB	-13.4	-12.9	-7.2	-3.0	1.9	15.2	15.0	14.7	6.9	0.9	-3.7	-9.5		
	2%	WB	-16.4	-16.7	-12.3	-6.7	0.0	6.9	9.6	9.6	4.6	-0.4	-7.2	-14.5		
		MCDB	-16.2	-16.5	-12.1	-6.4	0.6	8.8	11.5	11.5	5.2	0.0	-6.8	-14.2		
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	-18.8	-19.0	-16.7	-8.7	-1.2	4.7	7.7	8.0	3.7	-1.4	-9.8	-17.0		
		MCDB	-18.5	-18.8	-16.5	-8.3	-0.6	5.9	9.5	9.6	4.1	-0.9	-9.5	-16.7		
	10%	WB	-20.7	-21.0	-19.0	-10.7	-2.7	2.9	6.2	6.7	2.7	-2.4	-12.0	-19.0		
		MCDB	-20.5	-20.8	-18.8	-10.3	-2.1	3.7	7.7	7.9	3.2	-1.9	-11.6	-18.7		
		MDBR	5.6	5.5	5.5	6.1	4.4	3.0	3.9	3.8	2.5	4.2	5.4	5.5		
		MCDBR	7.7	7.4	7.8	7.1	4.5	6.8	8.2	7.4	3.5	3.1	6.8	8.1		
(36) Mean Daily Temperature Range	5% DB	MCWBR	7.6	7.3	7.7	6.9	4.2	5.4	6.3	5.7	3.2	3.0	6.7	8.0		
		MCDBR	7.7	7.3	7.7	7.0	4.5	6.3	8.1	7.2	3.4	2.8	6.8	8.1		
	5% WB	MCWBR	7.6	7.2	7.6	6.9	4.3	5.2	6.5	5.7	3.2	2.8	6.7	7.9		
		MCDBR	7.6	7.2	7.6	6.9	4.3	5.2	6.5	5.7	3.2	2.8	6.7	7.9		
(40) Clear-Sky Solar Irradiance	taub		N/A	0.168	0.261	0.299	0.325	0.318	0.344	0.335	0.279	0.193	N/A	N/A		
	taud		N/A	1.880	1.988	2.022	2.089	2.310	2.364	2.379	2.326	1.953	N/A	N/A		
	Ebn at Noon		0	194	588	755	803	839	794	735	619	133	0	0		
	Edh at Noon		0	13	64	103	115	97	89	74	48	10	0	0		
(44) All-Sky Solar Radiation	RadAvg		0.00	0.07	1.19	3.68	5.56	5.75	4.42	2.63	1.09	0.19	0.00	0.00		
	RadStd		0.00	0.01	0.07	0.21	0.34	0.62	0.44	0.28	0.09	0.01	0.00	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page